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CIS4394_Spring26

Agentic AI

02/04/26

Project Proposal: Budget-Based AI Travel Planning System

1. Problem statement & motivation

Traveling is something many people want to do, but for students and young adults, money is usually the biggest limitation. Most people do not always think about the destination as the first thing when planning a trip. Instead, they ask themselves questions like: How much can I spend? How many days do I have? Where can I realistically go? Unfortunately, most flight search tools today assume that the user already knows where they want to travel. This makes planning stressful and time-consuming, especially for people who are flexible and open to different destinations.

The goal of this project is to create an AI-powered travel system that answers the question "Where can I go with my budget?" Instead of forcing users to search one destination at a time, the system starts with the user's budget, dates, and home airport, and then recommends the best destinations available. The system also tracks flight prices, alerts users when prices change, and creates a short itinerary so users can quickly understand what their trip would look like.

The target users for this system are students, young professionals, and budget-conscious travelers who want simple, fast, and realistic travel suggestions. An agentic AI solution makes sense for this problem because travel planning involves many steps that depend on each other. The system must decide which destinations to consider, search flight prices, compare options, and then build an itinerary that fits the flight schedule. Using multiple AI agents allows each part of the problem to be handled independently while still working together toward one final result.

2. Use Cases & Scenarios

There are many different scenarios where this Agentic AI system might prove useful. For example, one of the most common scenarios is when a user wants to take a short trip but does not know where to go. With the AI system, the user enters their departure airport, travel dates, and a maximum budget. The system then searches multiple destinations, finds flights within the budget, ranks them by value, and presents the top options. This saves the user from manually searching dozens of locations. Another scenario is a weekend traveler who only has a few days off. In this case, the system avoids destinations with long travel times or inconvenient layovers. After selecting a destination, the system generates a short 2–3 day itinerary that fits around the arrival and departure times, making the trip feel realistic and manageable. One last scenario involves price tracking. After finding a destination they like, the user can set a price alert. The system periodically checks flight prices and notifies the user when the price changes. Instead of just showing a new price, the system explains the change in simple terms so the user can decide whether to book or wait.

3. Initial System Design

The system will be built using a multi-agent workflow, where each agent has a specific role. Below are some examples of the potential AI Agents that can be used:

- The Planner Agent decides which destinations to evaluate based on the user's budget and constraints.
- The Flight Search Agent retrieves live flight data and extracts key information such as price, stops, and travel time.
- The Destination Recommendation Agent ranks destinations using a value-based scoring system that considers price, number of stops, and total travel time.
- The Itinerary Agent creates a short daily plan that aligns with flight schedules.
- The Critic Agent reviews all outputs to ensure they are realistic and feasible.
- The Notification Agent handles price tracking and alerts.

In addition, this AI system will also use a flight search API to retrieve pricing data, a local database to store price history, and a web-based front-end that allows users to interact with the system easily.

4. Feasibility & Risks

One challenge of this project is dealing with flight price volatility, since prices change frequently. To manage this, the system will clearly label recommendations as estimates and avoid guaranteeing prices. Another challenge is avoiding unrealistic itineraries. This will be handled by the Critic Agent, which checks timing, buffers, and travel feasibility. API limitations and rate limits are also constraints, so results will be cached when possible. The project will not include ticket purchasing, only flight discovery and planning, which keeps the system manageable within the course timeline. The project will be developed throughout the semester, starting with a basic front-end, followed by flight search integration, multi-agent coordination, and finally evaluation and testing.

Conclusion

This project aims to create a budget-first travel planning system that makes trip discovery easier and more realistic for everyday users. By using a multi-agent AI approach, the system can independently search, evaluate, and refine travel options while still working toward a single goal. The final product demonstrates how agentic AI can be applied to a real-world problem in a practical and user-centered way.